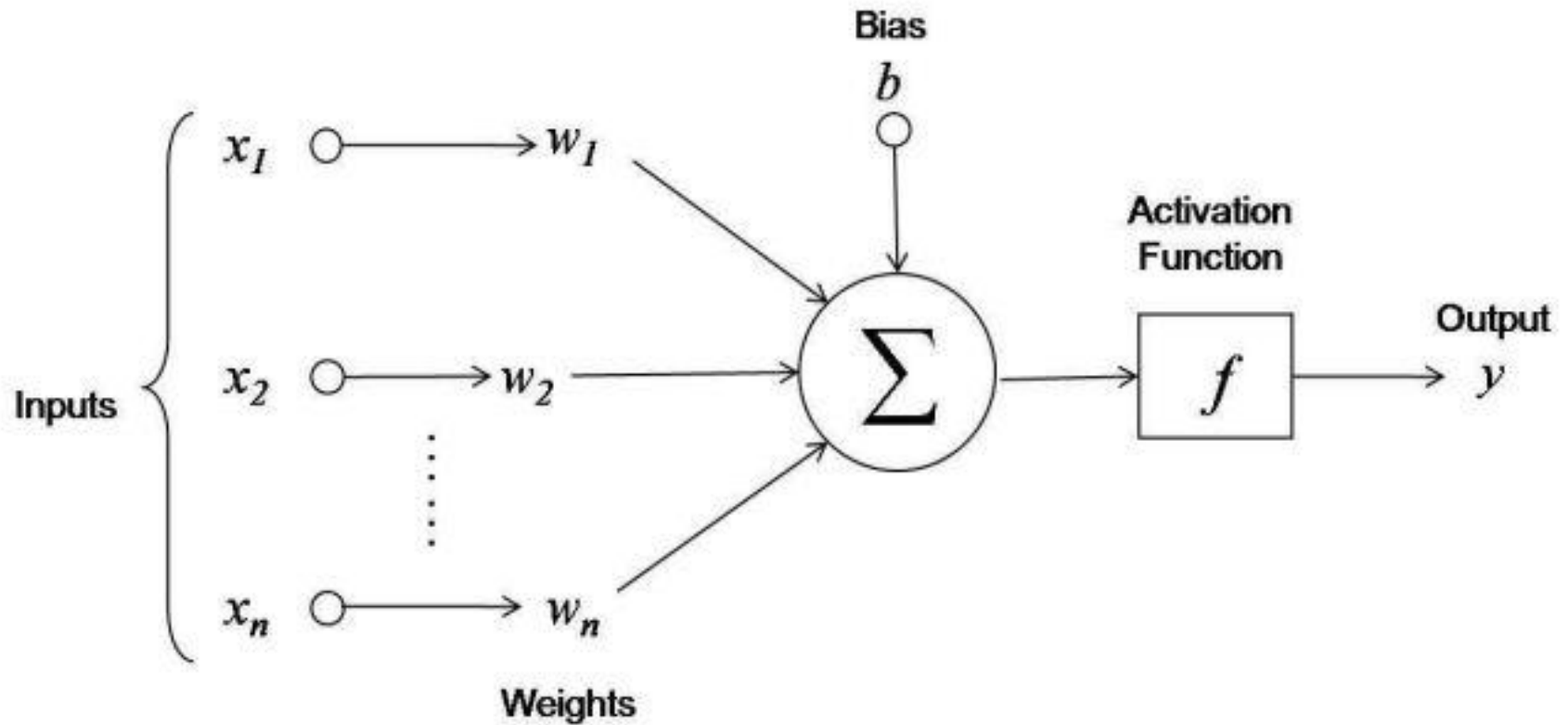


One Layer Neural Network



- We want to move the weight vector in the direction that decrease E
- $w_i = w_i + \Delta w_i$ $w = w + \Delta w$

$$\Delta w_i = -\eta \frac{\partial E}{\partial w_i}$$

$$\Delta \vec{w} = -\eta \nabla E[\vec{w}]$$

Gradient Descent

Gradient-Descent(*training_examples*, η)

Each training example is a pair of the form $\langle (x_1, \dots, x_n), t \rangle$ where $(x_1, \dots, x_n)$ is the vector of input values, and t is the target output value, η is the learning rate (e.g. 0.1)

- Initialize each w_i to some small random value
- Until the termination condition is met,
- Do
 - Initialize each Δw_i to zero
 - For each $\langle (x_1, \dots, x_n), t \rangle$ in *training_examples*
 - Do
 - Input the instance $(x_1, \dots, x_n)$ to the linear unit and compute the output o
 - For each linear unit weight w_i
 - Do
 - $\Delta w_i = \Delta w_i + \eta (t - o) x_i$
 - For each linear unit weight w_i
 - Do
 - $w_i = w_i + \Delta w_i$

$$\Delta w_i = \eta \sum_{d \in D} (t_d - o_d) x_{id}$$

Example: Solving XOR problem with Perceptron by 2 iterations

x_train =	0	0	targets =	0
	0	1		1
	1	0		1
	1	1		0

for bias term
expand data by adding one



x_train =	0	0	1	t =	0
	0	1	1		1
	1	0	1		1
	1	1	1		0

iter1:

$$w_0 = [0.1 \quad 0.2 \quad 0.3]$$

$$x = [1 \quad 0 \quad 1]$$

$$\text{sum} = w_0^T * x = 0.4$$

$$y = \begin{cases} 1 & \text{if sign(sum) > 0} \\ -1 & \text{if sign(sum) < 0} \end{cases}$$

$$y = 1$$

$$t = 1$$

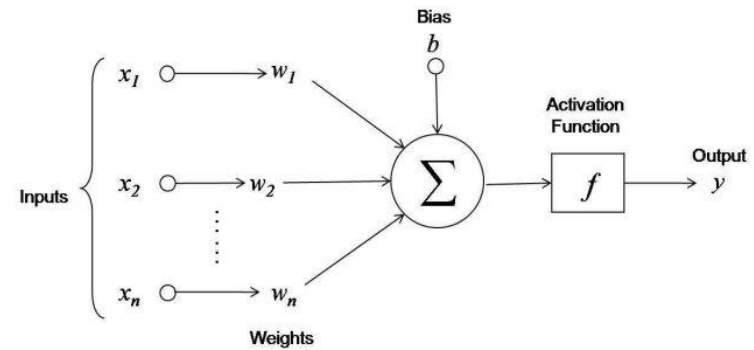
$$\text{error} = 0$$

$$\text{eta} = 0.01$$

$$\Delta w = \text{eta} * \text{error} * x$$

$$\Delta w = [0.0 \quad 0.0 \quad 0.0]$$

$$w_1 = w_0 + \Delta w = [0.1 \quad 0.2 \quad 0.3]$$



iter2:

$$w_1 = [0.1 \quad 0.2 \quad 0.3]$$

$$x = [1 \quad 1 \quad 1]$$

$$\text{sum} = w_1^T * x = 0.6$$

$$y = \begin{cases} 1 & \text{if sign(sum) > 0} \\ -1 & \text{if sign(sum) < 0} \end{cases}$$

$$y = 1$$

$$t = 0$$

$$\text{error} = t - y$$

$$\text{error} = -1$$

$$\text{eta} = 0.01$$

$$\Delta w = \text{eta} * \text{error} * x$$

$$\Delta w = [-0.01 \quad -0.01 \quad -0.01]$$

$$w_2 = w_1 + \Delta w = [0.09 \quad 0.19 \quad 0.29]$$